SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

K. J. Somaiya School of Engineering, Mumbai-77

Department of Computer Engineering

LAB CA

Course Name:	**Reinforcement Learning**	**Semester:**	**VII**
Date of Performance:	___ / ___ / _____	**DIV/ Batch No:**	
Student Name:		**Roll No:**	

Reinforcement Learning Lab CA – Mini Project Report

1. Title: _____

2. Problem Definition & RL Formulation

2.1 Problem Statement

Clearly describe the real-world problem addressed by the project.

2.2 Motivation

Explain why this problem is important and why **Reinforcement Learning** is suitable for solving it.

2.3 Objectives

List 2–4 specific objectives of the project.

2.4 RL Problem Formulation

RL Component	**Description**
Environment	
State (S)	
Action (A)	
Reward (R)	
Policy (π)	
Discount Factor (γ)	
Episode/Termination Condition	

2.5 Selected RL Algorithm

- Name of algorithm:
- Reason for selecting the algorithm:
- Brief description of the algorithm.

3. Implementation & Working of RL Agent

3.1 System Architecture / Workflow

Provide a flowchart or block diagram showing:

Environment → State → RL Agent → Action → Reward → State Update → Learning

3.2 Tools and Technologies

Mention:

- Programming language
- Libraries/frameworks
- Development environment
- Dataset/simulator, if applicable

3.3 Algorithm Implementation

Explain the major steps of the implementation.

3.4 Hyperparameters / Parameter Settings

Parameter	Value
Learning Rate (α)	
Discount Factor (γ)	
Exploration Rate (ε)	
Number of Episodes	
Other parameters	

3.5 Training Procedure

Explain how the RL agent was trained and how the policy/value function was updated.

3.6 Working Demonstration

Include **screenshots/output** showing that the RL agent is functioning.

4. Results, Analysis & Viva

4.1 Experimental Results

Present the results obtained after training.

4.2 Performance Metrics

Mention appropriate metrics, such as:

- Cumulative reward
- Average reward
- Success rate
- Number of steps
- Convergence
- Accuracy/performance improvement

4.3 Graphs and Visualization

Include relevant graphs such as:

- **Reward vs. Episode**
- **Success Rate vs. Episode**
- **Steps/Episode vs. Episode**

4.4 Result Analysis

Discuss:

- What did the agent learn?
- How did performance change during training?
- Did the agent converge?
- What observations can be made from the graphs?
- How effective is the learned policy?

4.5 Limitations

Mention the limitations of the proposed system.

4.6 Future Scope

Mention possible improvements and extensions.

4.7 Conclusion

Summarize the problem, approach, implementation, and major findings.

5. References

List books, research papers, websites, datasets, and documentation used.

Example:

1. Richard S. Sutton and Andrew G. Barto, *Reinforcement Learning: An Introduction*, 2nd Edition.
2. Relevant research papers.
3. Documentation of libraries/frameworks used.

Suggested Report Length

10–15 pages excluding references and appendix.

Appendix

- Source code / GitHub repository link
- Additional screenshots
- Sample outputs
- Important code snippets